



🎓 Education

Xinjiang Agricultural University · B.S. in Mathematics & Applied Mathematics Sep. 2023 – Jul. 2027
GPA 3.88/5.0 · **Coursework Top 1.5%** · **University-wide Rank 1st**
TOEFL 110 · Advisor: Prof. Hua Huang

12

Papers

9

First-author

2

In submission (Q1)

15

Projects (2 nat'l)

13

Awards (5 nat'l)

8

Patents & SW

Honors & affiliation: **National Scholarship**; Outstanding Volunteer (×9) & Outstanding Team, CYL Central Committee; Outstanding Student Pacesetter, Outstanding Student and Innovation & Entrepreneurship Star (3 consecutive years each), Science & Innovation Pioneer Scholarship, Outstanding Class Collective (as class monitor) — XJAU; Member, China Society for Industrial and Applied Mathematics.

🔬 Research & Projects

Xinjiang NSF General Program · Participant **Spatial–Spectral Fusion for Functional Modeling** **2026 – Present**

- **Situation** Hyperspectral imaging carries image and spectrum together, yet conventional pipelines process the two separately and concatenate them — breaking spatial structure while still facing band redundancy and the curse of dimensionality.
- **Action** I own image acquisition and preprocessing (9-member team, 10 months a year): ROI extraction, spectral correction, outlier removal and spline interpolation onto a common wavelength grid; then basis expansion turning discrete spectra into functional curves, with a basis-family and basis-number selection strategy settled by comparison.
- **Result** The pipeline now carries both representation routes — functional PCA and interval generalized PLS; the proposal further **seeded 3 student innovation projects** (1 regional, 2 university-level), whose sub-topics I lead-authored.

Provincial Project · Ongoing · PI **Shenji Miansuan — Sky-Ground Integrated Digital Cotton Field** **2026 – Present**

- **Situation** Pest scouting, water-and-fertiliser decisions and residual-film management on large cotton farms still rely on manual inspection and experience, with scattered sensing hardware and no digital layer turning data into decisions.
- **Action** **Project PI**, leading a four-person team and an industry partner; built a sky-ground monitoring network from ESP32 nine-axis environmental sensing and drone multispectral imaging, an improved YOLOv5 pest recogniser and a water–nitrogen–carbon–film coupled irrigation and fertilisation model, with edge-cloud collaboration and federated learning keeping each farm's data on site.
- **Result** **Two utility-model patents filed as first inventor** (plant-protection drones); advanced into the industry track of the 12th Xinjiang Innovation & Entrepreneurship Competition on a due-diligence recommendation.

National Project · Completed · PI **Small-Sample Time-Series Forecasting for Energy Economics** **2024 – 2025**

- **Situation** Energy-consumption series offer only a handful of annual points, where conventional forecasters overfit; grey models are viable here, but whether an integer accumulation order is optimal had never been tested.
- **Action** Lead-authored the design; re-engineered the kernel-regularized non-homogeneous grey model with conformable fractional accumulation (CFAKRNGM) and searched the continuous order by the Hiking Optimization Algorithm against a KRNGM baseline.
- **Result** MAPE fell from 0.0557 to 0.0522, but the optimum landed at $\alpha = 0.9998 / 0.9995$ — back at integer order, so **the fractional mechanism was never activated by the data**; the negative result is reported in full (co-first author).

Other 12 student innovation projects: **National (1)** — apple origin traceability (core member); **Provincial (4)** — Yunwen Jiangrong Hub (PI · Excellent), Apple SSC via FDA + image fusion (Excellent), Tongxi Zhixing visually-impaired navigation, hyperspectral multi-task detection; **University (7)** — nonlinear wave solutions of high-dimensional PDEs (PI · Excellent), cross-device spectral transfer for online rapid detection (PI), Zhitong smart cane (Excellent), Tianshan Pastoral Song (Excellent), non-traveling-wave solutions and localized excitations, AI pest & disease diagnosis, fruit & vegetable storage-quality forecasting; one of them recognized by the AI Ecosystem Group, Huawei Hangzhou Research Institute.

📖 Publications

- **Li P.L.**, Li Y.C., Li L.J., et al. *The label economics of calibration transfer: a five-benchmark, dual-modality evaluation of simple corrections versus deep and physics-informed representations*. Chemometrics and Intelligent Laboratory Systems. (SCI Q1; IF 3.8; Major Revision)
- **Li P.L.**, Yang S.Z., Li Y.C., Huang H.* *Functional regression modelling of soluble solids content in apple and hyperparameter selection with small calibration sets*. Journal of Xinjiang University (Natural Science Edition in Chinese and English). (Invited by editor; submitted)

- **Li P.L.** (presenter), Yang S.Z., Huang H.* *Upper bound of reference-value reliability and a sampling design table for near-infrared calibration of apple SSC*. The 11th National Conference on Near-Infrared Spectroscopy, China.
- **Li P.L.**, Zhang N. *Optimization of NIPT Testing Timing and Fetal Anomaly Determination Models*. International Journal of Public Health and Medical Research, 2026, 6(2): 1–8.
- **Li P.L.**, Bao F.C., Zhang J.Y., et al. *Periodic Solutions to a New Class of KdV Equations*. Advances in Applied Mathematics, 2025, 14(5): 23–28.
- **Li P.L.**, Zhang N. *Prediction of optimal NIPT timing and analysis of influencing factors of fetal Y chromosome concentration based on robust Bayesian ridge regression*. Academic Journal of Computing & Information Science, 2025, 8(10): 99–107.
- **Li P.L.**, Zhang N. *Study on the determination of fetal chromosomal abnormalities based on multi-model regression*. Academic Journal of Mathematical Sciences, 2025, 6(2): 121–129.
- **Li P.L.**, Zhang N. *Study on the effect of n-hexane insolubles on pyrolysis product yields based on statistical regression analysis*. Highlights in Science, Engineering and Technology, 2024, 116: 270–276.
- **Li P.L.**, Zhang N. *Research on Optimizing High-dimensional Data Model Based on Principal Component Analysis*. In: 2025 IEEE 3rd International Conference on Control, Electronics and Computer Technology (ICCECT), Jilin, China, 2025, pp. 998–1004.
- **Li P.L.**, Li J., Kadir A., et al. *Cultural Heritage and Rural Revitalization: Exploration and Reflections on China's Practice*. Smart Agriculture Guide, 2024, 4(21): 185–188.
- Zhang N., **Li P.L.**, Bao F.C., Liu X.Y. *Solitary Wave Solutions and Interaction Solutions of an Extended (3+1)-Dimensional Shallow Water Wave Equation*. Journal of Changchun Normal University, 2026, 45(4): 6–11.
- Jiao Q.T., **Li P.L.**, Mei T., et al. *ESP32-Based Multi-Sensor Near- & Far-Range Water Quality Monitoring System*. Internet of Things Technologies, 2026, 16(2): 9–12, 16.
- **Utility-Model Patents (4, as first inventor)**: Section-Connection Component for a White Cane · Portable Ultrasonic Detection Device · Pesticide Spraying Drone · Foldable Anti-Collision Spraying Boom for Drones.
- **Software Copyrights (4, as first author)**: Intelligent White Cane Vision Recognition System · Smart Cane Data Collection & Monitoring System · User Behavior Analysis and Recommendation System · Big-Data-Driven Market Analysis and Forecasting System.

🏆 Competitions & Awards

China International College Students' Innovation Competition (2024)	National · Bronze · Participant
11th National Undergraduate Statistical Modeling Contest	National · Third Prize · Team Lead
“Chia Tai Cup” National Market Survey & Analysis Contest (15th & 16th)	National · Third Prize × 2 · Team Lead
“DataBee Cup” National Interview Survey Contest	National · Third Prize · Team Lead
China International College Students' Innovation Competition, Xinjiang Division (2025)	Provincial · Bronze · Team Lead
National Mathematical Modeling Contest, Xinjiang Division (2024)	Provincial · Second Prize · Team Lead
10th National Undergraduate Statistical Modeling Contest, Xinjiang Division	Provincial · Second Prize · Team Lead
14th China Innovation & Entrepreneurship Competition, Xinjiang Division	Provincial · Third Prize · Team Lead
Xinjiang Division Information Literacy Contest (2024)	Provincial · Merit Award · Team Lead

🌱 Student Experience

Class Monitor , Mathematics 2302, School of Mathematics & Physics, Xinjiang Agricultural University	Sep. 2023 – Jul. 2027
Class Monitor , 19th Youth Marxist Training Program, Xinjiang Agricultural University	May 2025 – May 2026
Head , Volunteer Association, School of Mathematics & Physics	Sep. 2024 – Nov. 2025
• Founded the 20-member “Shield of the Future” volunteer team; took part in 11 public-welfare programs organized by the CYL Central Committee and the China Young Volunteers Association; the work was covered by <i>China Daily</i>, <i>People's Daily</i> and other outlets.	